

Section-A. MCQs

1. (b) 8
2. (b) contains all elements under consideration
3. (a) $4/5$
4. (c) many - one relation
5. (b) $1/2$
6. (c) 60
7. (b) $A \cap B = \emptyset$
8. (c) Transitive

Section B - Very short Answers

9. A set is a well-defined collection of objects.

Example: $\{1, 2, 3\}$.

10. Domain = $\mathbb{R}$, Range = $[0, \infty)$.

11. $\sin 45^\circ \cos 45^\circ = 1/2$

12. $5P2 = 20$.

13. Power set elements = 16.

14. One-one: $F(x) = x$, Many-one: $F(x) = x^2$

Section C - short Answers

15. Identity proved, minor algebraic steps skipped.

16. $A \cup B = \{1, 2, 3, 4, 5, 6\}$. $A \cap B = \{3, 4\}$.

17. $5P3 = 60$.

18. Relation is reflexive, symmetric and transitive.

Section D - Long Answer

19. $(1 + \tan^2 \theta)(1 + \sin^2 \theta) = \sec^2 \theta (1 + \sin^2 \theta)$ Identity proved with brief explanation.